



# Phonological Alternation Learning is Facilitated by Relative Frequency Cues

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## 1. Alternations and Decomposition

**MAIN QUESTION:** Is there a link between lexical frequency distribution and phonological representations?

Speakers are attuned to differences in **lexical decomposition** and may rely on them to learn idiosyncrasies in morphological productivity [4, 5], phonotactics [6], and phonological alternations [2, 3, 12]

**Decomposition:** Abstract lexical parsing process part of lexical access; predicted using **token relative frequency** [4, 5, 6]

Decomposed parse vs Whole-word parse  
/move/ + /-ment/ vs /movement/  
Stem freq. > affixed freq. vs Affixed freq. > stem freq.

**THIS STUDY:** Strengthens potential link between decomposition and phonological representations by investigating effects of relative frequency on varied phonological rule application

(1) English Stress Shift: Word prominence shifts rightward when suffixes are added [1, 8]

a. párent + al → [parént-al]    b. párent + hood → [párent-hood]  
Application suffix ☺    Non-application suffix ☹

Previous analyses claim speakers have internalized application and non-application categories which derive correct phonological patterns [1, 8]

**What information do learners use to establish these categories?**

**Hypothesis:** Speakers rely on cues to lexical decomposition (e.g. relative frequency) to accurately learn phonological patterns with varied application

## 3. Procedure

**Artificial language learning study (3 phases):**

**1. Learning:** Shows examples of how the artificial language works

**2. Verification:** Re-presents learning stimuli and asks participants to verify correct pronunciations

**3. Test:** Asks participants to generalize pattern to novel items

**THIS STUDY:** Learning phase shows meanings of stems and suffixes, phonological rule application, and artificial frequency distribution [7, 10]

Stem and suffixed forms presented in **particular relative frequency ratio** which encourages or discourages decomposition for a given suffix

Decomposed parse vs Whole-word parse  
Stem freq. > affixed freq. vs Affixed freq. > stem freq.

**Figure 1: Experimental Conditions for Both Suffix Types:**

|                               |                  |                  |                    |
|-------------------------------|------------------|------------------|--------------------|
| <b>Non-application suffix</b> | No frequency cue | Decomposed parse | Whole-word parse   |
|                               |                  |                  |                    |
| <b>Application suffix</b>     | No frequency cue | Whole-word parse | Decomposed parse   |
|                               | <b>Baseline</b>  | <b>Congruent</b> | <b>Incongruent</b> |

**Decomposed parse** strengthens separate stem and suffix representations (e.g. /move/ + /ment/), making the **non-application pattern** easier to learn

**Whole-word parse** promotes combined stem-suffix representation (e.g. /movement/) making the **non-application pattern** harder to learn

**Prediction:** Test phase response accuracy in **congruent condition** > **incongruent condition**

## 2. Participants and Stimuli

Artificial language learning (ALL) study

180 native English-speakers participated in PCIBex [11] experiment, recruited via Prolific [9], excluded 13 → **167 total participants**

Nonsense word stems (CVCVC) and 2 nonce suffixes (VCV)

Stem-final voicing assimilation rule applies with one suffix but not the other  
[porap] + [-iso] → [porab-iso] (**assimilation applies – application suffix**)  
[porap] + [-ifa] → [porap-ifa] (**no assimilation – non-application suffix**)

All stimuli associated with meanings

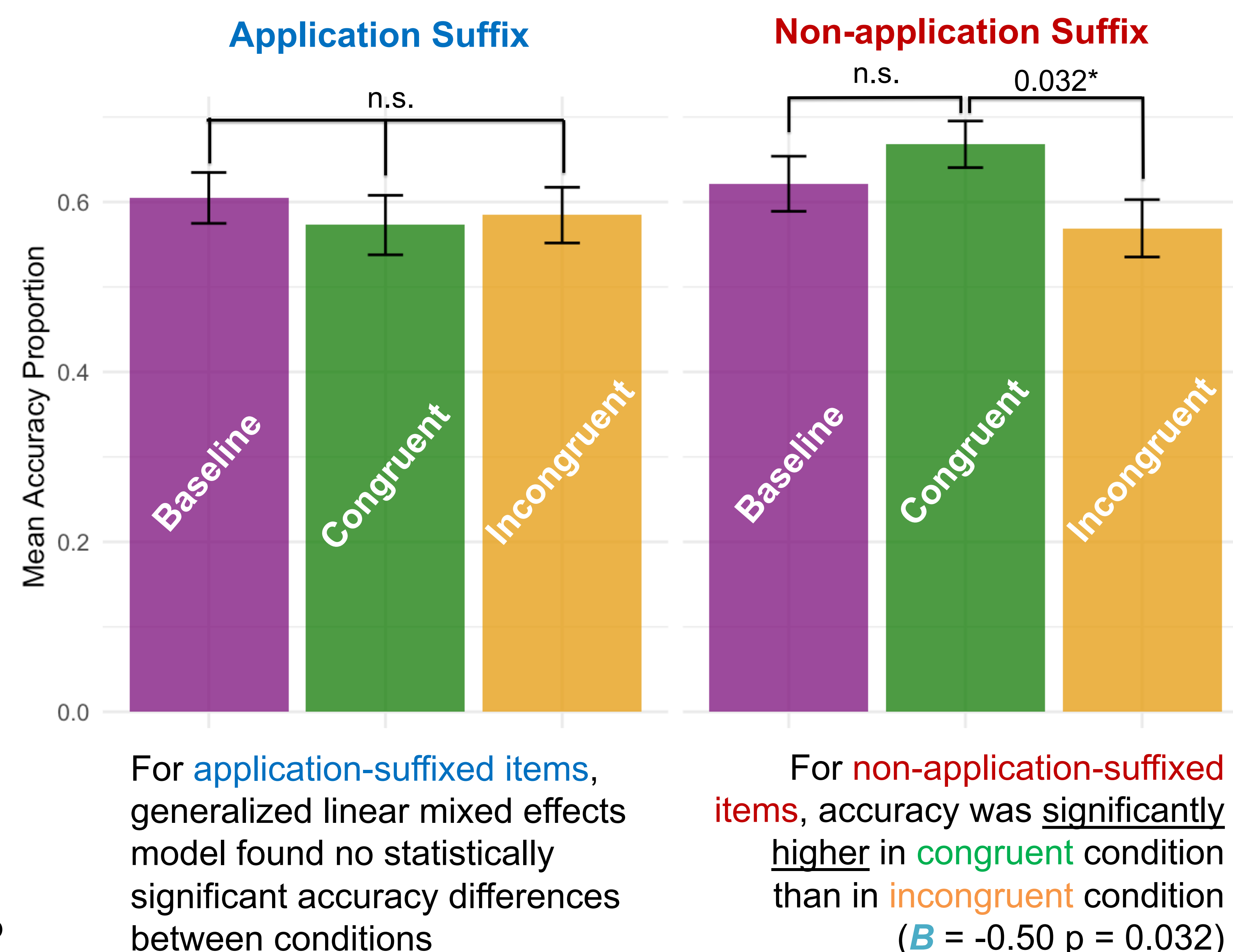
Stem = objects and animals, suffix 1 = 'small', suffix 2 = 'big'

Suffix meanings and application/nonapplication counterbalanced

138 total trials (88 learning, 20 verification, 30 test)

## 4. Results

**Figure 2: Accuracy for Each Suffix Type Between Conditions:**



## 5. Conclusions

**STUDY GOAL:** Find evidence that decomposition (indicated by relative frequency cues) facilitates learning of application and non-application suffixes in phonology

Results showed that in the **congruent** condition, learning of the non-application pattern was *facilitated* compared to the **incongruent** condition

Suggests that lexical parsing processes like decomposition play an active role in helping speakers learn which forms should/should not undergo phonological alternations

Supports phonological theories which incorporate distributional lexical information into phonological representations

Interestingly, decomposition cues did not show significant learning differences for application pattern across conditions  
- Is productivity (type frequency) more relevant here?

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